

## NETWORKS AND COMPLEXITY

### Exercise Sheet 17: Coarse-graining

*This is an exercise sheet from the forthcoming book *Networks and Complexity*.  
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

#### **Ex 17.1: Simple timescale renormalization [Lvl. 1]**

Change the timescale to make the parameter  $k$  disappear from the system

$$\dot{x} = k(x - x^2)$$

#### **Ex 17.2: Codimension example [Lvl. 2]**

Consider the system

$$\dot{x} = f(x, a, b) = a - bx^2.$$

For  $b > 0$  find the steady states and their bifurcation points. Plot the bifurcation points in a 2-parameter bifurcation diagram, where the axes are the parameters  $a$  and  $b$ .

#### **Ex 17.3: Simple conservation law [Lvl. 2]**

If you did the exercises of the previous chapters you might be able to guess the conservation law found in the dynamical system

$$\dot{x} = y \qquad \dot{y} = -x$$

Can you also find the conservation law mathematically?

#### **Ex 17.4: Linear manifold example [Lvl. 2]**

Use eigendecomposition to find the invariant manifolds of the steady state in the following linear system:

$$\begin{aligned}\dot{x} &= 2x + 4y \\ \dot{y} &= x - y,\end{aligned}$$

#### **Ex 17.5: Manifold approximation [Lvl. 2]**

Find a second order (quadratic) approximation to the invariant manifolds of the steady state at  $(0, 0)$  in the following system:

$$\begin{aligned}\dot{x} &= 2x + 4y - x^2 \\ \dot{y} &= x - y + xy\end{aligned}$$

#### **Ex 17.6: Abstract quasi-steady state [Lvl. 2]**

Use a quasi-steady state approximation to analyze the dynamics of the system

$$\begin{aligned}\dot{x} &= -x^3 + x + y \\ \dot{y} &= -\alpha x\end{aligned}$$

where  $0 < \alpha \ll 1$ . (The result may be surprising).

**Ex 17.7: Reproduction, reduction and cats [Lvl. 3]**

Consider a population of male cats  $M$ , female cats that are not pregnant or raising kits  $F$ , and female cats that are pregnant or raising kits  $P$ . Female cats in the class  $F$  become pregnant through contact with males. Females in the class  $P$  return to class  $F$  when their offspring reach adulthood.

- Write a mass action model for this system.
- Assume that becoming pregnant is a much faster process than raising kits. Then, use the Tikhonov-Fenichel reduction to formulate a model for the total number of males  $M$  and females  $X = P + F$ .

**Ex 17.8: Lotka-Volterra [Lvl. 3]**

The Lotka-Volterra model describes the interaction of a prey population  $X$  and a predator population  $Y$  by

$$\begin{aligned}\dot{X} &= aX - bXY \\ \dot{Y} &= cXY - mY\end{aligned}$$

Find the conservation law hiding in this model.

**Ex 17.9: Holling's type-II response [Lvl. 3]**

To develop a better model of predator-prey dynamics, we consider the number of prey  $X$ , predators searching for prey  $S$ , and predators currently handling prey  $H$ , which change in time according to

$$\begin{aligned}\dot{X} &= aX - rXS \\ \dot{S} &= -rXS + bH + cH - mS \\ \dot{H} &= +rXS - bH - mH\end{aligned}$$

where predator reproduction  $cH$  and mortality  $mS, mH$  are slow.

- Ignore the equation for prey for now. Use Tikhonov-Fenichel reduction to find the one-dimensional slow subsystem of the predator dynamics.
- Call the slow predator variable  $Y$  and write a closed model consisting of ODEs for  $X$  and  $Y$ . Compare the result with the Lotka-Volterra model.

**Ex 17.10: Self-organized criticality [Lvl. 4]**

Let's consider once again the mean-field SIS model,  $\dot{I} = pzI(1 - I) - I$ . But this time we take two additional processes into account: The agents create new connections in the network at constant rate. If there is an ongoing epidemic people break links proportional to the number of infected, such that  $\dot{z} = a - bI$

- Compute the steady state of the system as a function of  $p$  is there still an epidemic threshold?
- Now consider the case where  $b = \varepsilon$  and  $a = \varepsilon^2$ . What happens in the limit  $\varepsilon \rightarrow 0$ ?

**Ex 17.11: SEIS center manifold [Lvl. 4]**

The SEIS model is an epidemic model in which contact between a susceptible and an infectious agent causes the susceptible agent to enter an exposed (E) state, where it carries the disease but

isn't infectious yet. Exposed agents then progress to the infectious (I) state before eventually recovering back to the susceptible state. The SEIS model is described by the equations

$$\begin{aligned}\dot{E} &= pI(1 - E - I) - E \\ \dot{I} &= E - I.\end{aligned}$$

Our goal in this exercise is to decide whether a discontinuous transition takes place at the epidemic threshold, without computing nontrivial steady states.

- a) Find the epidemic threshold, by analyzing the stability of the trivial state.
- b) Compute the eigenvectors in the bifurcation point. Define new coordinates  $x$  and  $y$  such that  $x$  is the coordinate in the direction of the eigenvector with the zero eigenvalue and  $y$  is the coordinate in the direction of the other eigenvalue. (Keep  $p$  in the equations for now.)
- c) Introduce a new parameter  $\rho = p - 1$ . Use the ansatz  $y = h(x, \rho) = ax + O(2)$  to find a (very simple) approximation for the center manifold.
- d) Constrain your dynamics to the center manifold (substitute your  $y$  into  $\dot{x}$ ). Compare the result to the normal form of the transcritical bifurcation. Is the bifurcation in the SEIS model *supercritical* as in the SIS model or *subcritical* as in the aSIS model with strong rewiring (see Ex. 16.11)?
- e) Perhaps the linear approximation for  $y$  was a bit reckless. Try again with the ansatz  $y = bx^2 + cx\rho + O(3)$ .

### Ex 17.12: SIS Closure complications [Lvl. 5]

Let's consider these two equations from the pair approximation of the SIS model

$$\begin{aligned}[\dot{S}S] &= r[SI] - 2p[SS][SI]/[S] \\ [\dot{I}I] &= p[SI] + p[SI]^2/[S] - 2r[II].\end{aligned}$$

with  $[SI] = z/2 - [SS] - [II]$ . Assume that the link dynamics described by these equations is much faster than the dynamics of  $[S]$  (not shown). Solve the link equations to find an expression that gives you  $[SI]$  as a function of  $[S]$ . Does the resulting  $[SI]$  behave like you would expect it to? Is there a way to fix this?